

Module-01**RBT Level-L1, L2****Image Fundamentals and Image Processing Techniques:**

Introduction to image processing, Basic processing: histogram operations, convolution, derivatives. Image filtering, Edge and corner detection, Morphological operations, Histogram equalization, thresholding, Overview of OpenCV and Vision tools, Ethical Concerns and Challenges in Computer Vision.

Introduction to image processing

Digital Image Processing means processing digital image by means of a digital computer. Digital image processing is the use of algorithms and mathematical models to process and analyze digital images. The goal of digital image processing is to enhance the quality of images, extract meaningful information from images, and automate image-based tasks.

The basic steps involved in digital image processing are:

- **Image acquisition:** This involves capturing an image using a digital camera or scanner, or importing an existing image into a computer.
- **Image enhancement:** This involves improving the visual quality of an image, such as increasing contrast, reducing noise, and removing artifacts.
- **Image restoration:** This involves removing degradation from an image, such as blurring, noise, and distortion.
- **Image segmentation:** This involves dividing an image into regions or segments, each of which corresponds to a specific object or feature in the image.
- **Image representation and description:** This involves representing an image in a way that can be analyzed and manipulated by a computer, and describing the features of an image in a compact and meaningful way.
- **Image analysis:** This involves using algorithms and mathematical models to extract information from an image, such as recognizing objects, detecting patterns, and quantifying features.
- **Image synthesis and compression:** This involves generating new images or compressing existing images to reduce storage and transmission requirements.

Image processing mainly include the following steps:

- Importing the image via image acquisition tools;
- Analyzing and manipulating the image;
- Output in which result can be altered image or a report which is based on analyzing that image.

What is an image?

An image is defined as a two-dimensional function, $F(x,y)$, where x and y are spatial coordinates, and the amplitude of F at any pair of coordinates (x,y) is called the intensity of that image at that point. When x,y , and amplitude values of F are finite, we call it a digital image.

Types of an image

- **Binary Image-** The binary image as its name suggests, contain only two pixel elements i.e 0 & 1, where 0 refers to black and 1 refers to white. This image is also known as Monochrome.
- **Black and White Image-** The image which consist of only black and white color is called BLACK AND WHITE IMAGE.
- **8 Bit Color Format-** It is the most famous image format. It has 256 different shades of colors in it and commonly known as Grayscale Image. In this format, 0 stands for Black, and 255 stands for white, and 127 stands for gray.
- **16 Bit Color Format-** It is a color image format. It has 65,536 different colors in it. It is also known as High Color Format. In this format the distribution of color is not as same as Grayscale image.

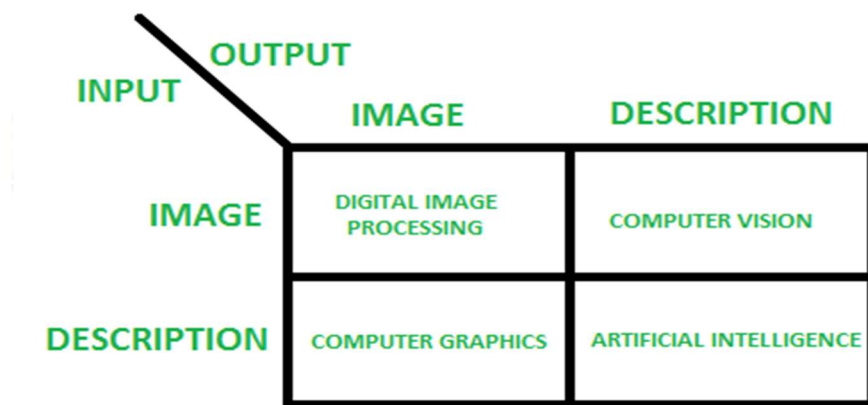
A 16 bit format is actually divided into three further formats which are Red, Green and Blue. That famous RGB format.

PHASES OF IMAGE PROCESSING:

- **Acquisition-** It could be as simple as being given an image which is in digital form. The main work involves:
 - ❖ Scaling
 - ❖ Color conversion (RGB to Gray or vice-versa)
- **Image Enhancement-** It is amongst the simplest and most appealing in areas of Image Processing it is also used to extract some hidden details from an image and is subjective.

- **Image Restoration-** It also deals with appealing of an image but it is objective (Restoration is based on mathematical or probabilistic model or image degradation).
- **Color Image Processing-** It deals with pseudocolor and full color image processing color models are applicable to digital image processing.
- **Wavelets And Multi-Resolution Processing-** It is foundation of representing images in various degrees.
- **Image Compression-**It involves in developing some functions to perform this operation. It mainly deals with image size or resolution.
- **Morphological Processing-**It deals with tools for extracting image components that are useful in the representation & description of shape.
- **Segmentation Procedure-**It includes partitioning an image into its constituent parts or objects. Autonomous segmentation is the most difficult task in Image Processing.
- **Representation & Description-**It follows output of segmentation stage, choosing a representation is only the part of solution for transforming raw data into processed data.
- **Object Detection and Recognition-**It is a process that assigns a label to an object based on its descriptor.

Overlapping Fields with Image Processing



According to block 1: if input is an image and we get out image as a output, then it is termed as Digital Image Processing.

According to block 2: if input is an image and we get some kind of information or description as a output, then it is termed as Computer Vision.

According to block 3: if input is some description or code and we get image as an output, then it is termed as Computer Graphics.

According to block 4: if input is description or some keywords or some code and we get description or some keywords as a output, then it is termed as Artificial Intelligence

Advantages of Digital Image Processing:

Improved image quality: Digital image processing algorithms can improve the visual quality of images, making them clearer, sharper, and more informative.

Automated image-based tasks: Digital image processing can automate many image-based tasks, such as object recognition, pattern detection, and measurement.

Increased efficiency: Digital image processing algorithms can process images much faster than humans, making it possible to analyze large amounts of data in a short amount of time.

Increased accuracy: Digital image processing algorithms can provide more accurate results than humans, especially for tasks that require precise measurements or quantitative analysis.

Disadvantages of Digital Image Processing:

High computational cost: Some digital image processing algorithms are computationally intensive and require significant computational resources.

Limited interpretability: Some digital image processing algorithms may produce results that are difficult for humans to interpret, especially for complex or sophisticated algorithms.

Dependence on quality of input: The quality of the output of digital image processing algorithms is highly dependent on the quality of the input images. Poor quality input images can result in poor quality output.

Limitations of algorithms: Digital image processing algorithms have limitations, such as the difficulty of recognizing objects in cluttered or poorly lit scenes, or the inability to recognize objects with significant deformations or occlusions.

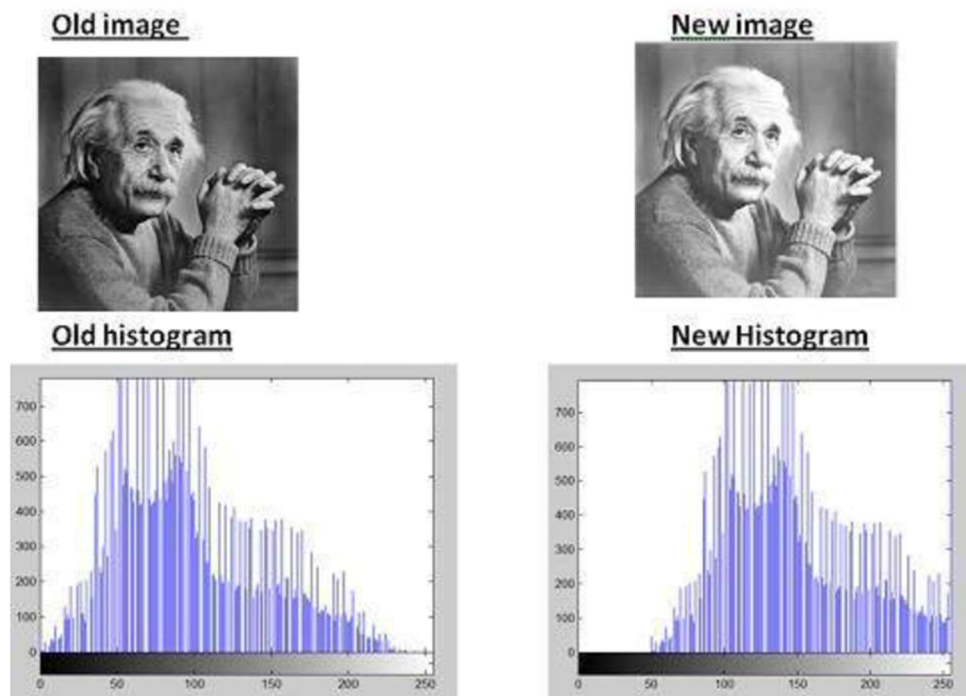
Dependence on good training data: The performance of many digital image processing algorithms is dependent on the quality of the training data used to

develop the algorithms. Poor quality training data can result in poor performance of the algorithm.

Histogram operations

Histogram is nothing but a graph that shows frequency of occurrence of data.

Let's compare these two images and their histograms to see that what change have to got.



On the **y** axis of this histogram are the frequency or count. **x** axis, we have gray level values.

Histogram Equalization: Histogram equalization is used to enhance contrast. Now we will perform histogram equalization to it.

- ❖ **Probability Mass Function (PMF)**
- ❖ **Cumulative Distributive Function(CDF)**

Probability Mass Function (PMF)

The probability mass function (PMF) of a random variable **X** is a function which specifies the probability of obtaining a number **X(ξ) = a**.

Probability Mass Function Definition: It is represented as $p_X(a) = P[X = a]$.
 where, **Function X:** the random variable which translates words to numbers
Function p_X : the mapping from event $\{X = a\}$ to a probability

Example 1: Probability mass function is given by: $f(x) = ax^2$ for $x = 0, 1, 2$ then, find the value of a .

Solution: $p_X(a) = P[X = a] = 1$
 $P(0) + P(1) + P(2) = 1$
 $a \times 0^2 + a \times 1^2 + a \times 2^2 = 1$
 $a \times (0^2 + 1^2 + 2^2) = 1$
 $a \times (0 + 1 + 4) = 1$
 $5a = 1$
 $a = 1/5$

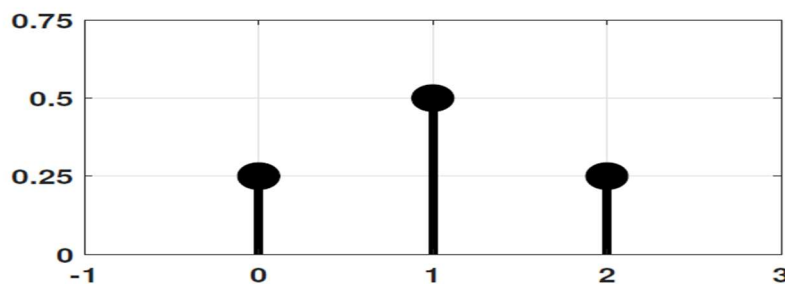
Example 2: Table below represents the PMF for a random variable X . Find the value of p .

X	2	4	6
$P(X = x)$	p	$p + 0.5$	0.2

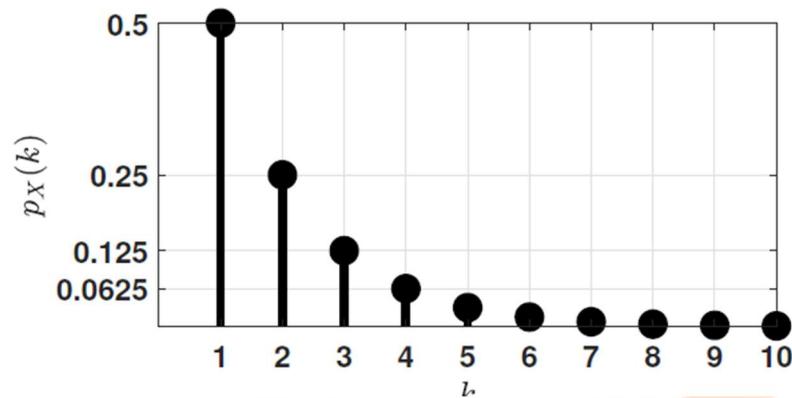
Solution: $p_X(a) = P[X = a] = 1$
 $p + p + 0.5 + 0.2 = 1$
 $2p + 0.7 = 1$
 $2p = 1 - 0.7$
 $2p = 0.3$
 $p = 0.3 / 2$
 $p = 0.15$

Example 3: Flip a coin twice. Define X = number of heads. Then the probability mass function is

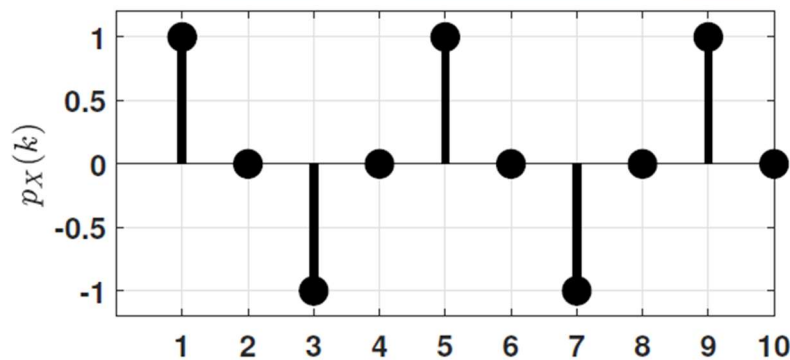
$p_X(0) = P[X = 0] = P[\{\text{"TT"}\}] = 1/4$
 $p_X(1) = P[X = 1] = P[\{\text{"TH"}, \text{"HT"}\}] = 1/2$
 $p_X(2) = P[X = 2] = P[\{\text{"HH"}\}] = 1/4$



Example 4: Let X be a random variable with PMF, $p_X(k) = c(1/2)^k$, $k = 1, 2, \dots$. Find c .



Example 4: Let X be a random variable with PMF $p_X(k) = c \sin(\pi/2)^k$, for $k = 0, 1, 2, \dots$. Find c .



CDF: Our next step involves calculation of CDF (cumulative distributive function). Again if you donot know how to calculate CDF , please visit our tutorial of CDF calculation.

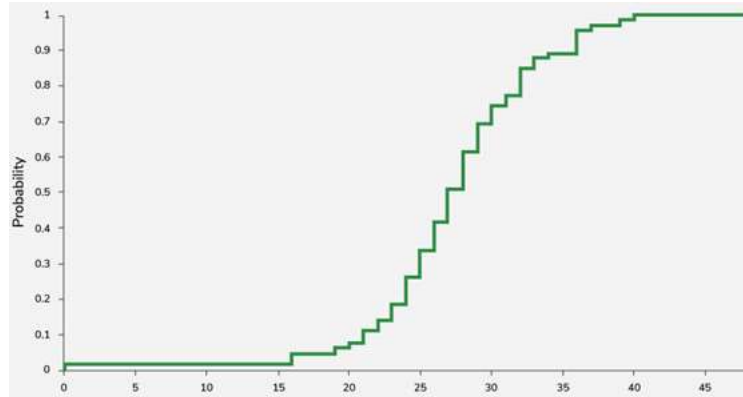
Cumulative Distributive Function(CDF)

Cumulative Distribution Function (CDF) of a random variable is a mathematical function that provides the probability that the variable will take a value less than or equal to a particular number.

The CDF starts at **0** for the smallest possible value of **X** and increases to **1** as **x** approaches the largest possible value of **X**.

The Cumulative Distribution Function $F(x)$ of the random variable X is defined as: $F(x) = P(X \leq x)$

Graphical Representation of Cumulative Distribution Function



How to Calculate Cumulative Distribution Function?

Step 1. Identify the Distribution: The Determine whether the random variable follows a discrete or continuous distribution.

Step 2. Determine the PDF: For continuous distributions find the Probability Density Function (PDF) $f(x)$.

Step 3. Integrate the PDF: The Integrate the PDF from the $-\infty$ to x to find the CDF:

$$F(x) = \int_{-\infty}^x f(t) dt$$

Step 4. Sum the Probabilities: For discrete distributions sum the probabilities for the all values less than or equal to the x .

Using Probability Density Function (PDF) to Find CDF

For continuous random variables the CDF $F(x)$ is derived from the PDF $f(x)$ by the integrating:

$$F(x) = \int_{-\infty}^x f(t) dt$$

Example 1: Consider a random variable X representing the number of the heads obtained in the two-coin flips. Calculate the CDF $F(x)$ of X .

The possible values of X are 0, 1 and 2. The probabilities are:

$$P(X=0)=1/4(\text{no heads})$$

$$P(X=1)=1/2(\text{one head})$$

$$P(X=2)=1/4(\text{two heads})$$

The CDF $F(x)$ is:

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{4} & \text{if } 0 \leq x < 1 \\ \frac{3}{4} & \text{if } 1 \leq x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$

Example 2: Given the CDF $F(x)=1-e^{-x}$ for $x \geq 0$ find the probability that X lies between the 1 and 2.

The probability that X lies between the 1 and 2 is:

$$P(1 \leq X \leq 2) = F(2) - F(1)$$

Calculate:

$$F(2) = 1 - e^{-2}$$

$$F(1) = 1 - e^{-1}$$

$$P(1 \leq X \leq 2) = (1 - e^{-2}) - (1 - e^{-1}) \\ = e^{-1} - e^{-2}$$

Example 3: Given a PDF $f(x) = \frac{3}{4}(1-x^2)$ for the $-1 \leq x \leq 1$ find the CDF $F(x)$.

Integrate the PDF to find the CDF:

$$F(x) = \int_{-\infty}^x f(t) dt = \int_{-1}^x \frac{3}{4}(1 - t^2) dt$$

Solving the integral:

$$F(x) = \frac{3}{4} \left[t - \frac{t^3}{3} \right]_{-1}^x = \frac{3}{4} \left(x - \frac{x^3}{3} + \frac{2}{3} \right)$$

The CDF ($F(x)$) is:

$$F(x) = \begin{cases} 0 & \text{if } x < -1 \\ \frac{3}{4} \left(x - \frac{x^3}{3} + \frac{2}{3} \right) & \text{if } -1 \leq x \leq 1 \\ 1 & \text{if } x > 1 \end{cases}$$

Convolution: A mathematical operation that modifies an image by applying a smaller matrix, called a kernel or filter, to the original image.



It can be mathematically represented as : $g(x,y) = h(x,y) * f(x,y)$

- ❖ convolution operator(*)
- ❖ The $h(x,y)$ is the mask or filter.

What is mask?

- ❖ Mask is also a signal. It can be represented by a two-dimensional matrix. The mask is usually of the order of 1x1, 3x3, 5x5, 7x7.
- ❖ A mask should always be in odd number, because otherwise you cannot find the mid of the mask.

How to perform convolution?

- ❖ Flip the mask (horizontally and vertically) only once
- ❖ Slide the mask onto the image.
- ❖ Multiply the corresponding elements and then add them
- ❖ Repeat this procedure until all values of the image has been calculated.

Example of convolution:

$$f * h = \sum_k \sum_l f(k,l)h(i-k, j-l)$$

f = Image
 h = Kernel

f

f_1	f_2	f_3
f_4	f_5	f_6
f_7	f_8	f_9

 $\otimes$

h_9	h_8	h_7
h_6	h_5	h_4
h_3	h_2	h_1

 $\rightarrow$

h

h_1	h_2	h_3
h_4	h_5	h_6
h_7	h_8	h_9

 $\xrightarrow{X-flip}$

h_7	h_8	h_9
h_4	h_5	h_6
h_1	h_2	h_3

 $\xrightarrow{Y-flip}$

h_9	h_8	h_7
h_6	h_5	h_4
h_3	h_2	h_1

$f * h = f_1h_9 + f_2h_8 + f_3h_7$
 $+ f_4h_6 + f_5h_5 + f_6h_4$
 $+ f_7h_3 + f_8h_2 + f_9h_1$

1	2	3
4	5	6
7	8	9

Step 1 is to flip the mask. **Flipping the mask horizontally**

3	2	1
6	5	4
9	8	7

Flipping the mask vertically

9	8	7
6	5	4
3	2	1

Image : Lets consider an image to be like this

2	4	6
8	10	12
14	16	18

Convolution

9	8	7		
6	2	5	4	4
3	8	2	10	1
	14		16	
				6
				12
				18

- ❖ The box in **red** color is the mask, and
- ❖ The values in the **orange** are the values of the mask.
- ❖ The **black** color box and values belong to the image.

Now for the first pixel of the image, the value will be calculated as

$$\begin{aligned}
 \text{First pixel} &= (5*2) + (4*4) + (2*8) + (1*10) \\
 &= 10 + 16 + 16 + 10 \\
 &= 52
 \end{aligned}$$

Derivatives.

Derivative is defined as a rate of change.

Discrete Derivative Finite Distance

Backward difference: {-1 1}

$$\frac{df}{dx} = f(x) - f(x-1) = f'(x)$$

Forward difference: {1 -1}

$$\frac{df}{dx} = f(x) - f(x+1) = f'(x)$$

Central difference: {-1 0 1}

$$\frac{df}{dx} = f(x+1) - f(x-1) = f'(x)$$

Edge detection

Edge detection is the technique used to identify the regions in the image where the brightness of the image changes sharply.

To detect an edge using the forward operator we use given below steps:

Steps:

- ❖ Read the image.
- ❖ Convert into grayscale if it is colored.
- ❖ Convert into the double format.
- ❖ Define the mask or filter.
- ❖ Detect the edges along X-axis.
- ❖ Detect the edges along Y-axis.
- ❖ Combine the edges detected along X and Y axes.
- ❖ Display all the images.

Forward operator [1 -1 0]:**% MATLAB code for Forward operator edge detection**

```

k=imread("logo.png");
k=rgb2gray(k);
k1=double(k);
forward_msk=[1 -1 0];
kx=conv2(k1, forward_msk, 'same');
ky=conv2(k1, forward_msk, 'same');
ked=sqrt(kx.^2 + ky.^2);

```

%display the images.

```
imtool(k,[]);
```

%display the edge detection along x-axis.

```

imtool(kx,[]);
imtool(abs(kx), []);

```

%display the edge detection along y-axis.

```

imtool(ky,[]);
imtool(abs(ky),[]);

```

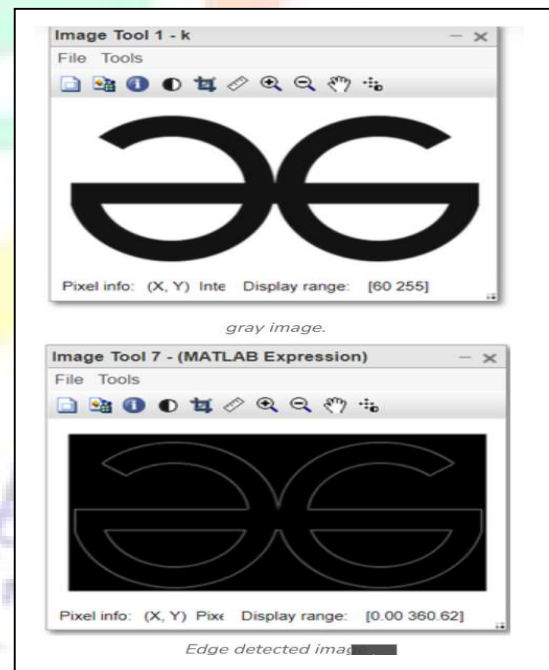
%display the full edge detection.

```

imtool(ked,[]);
imtool(abs(ked),[]);

```

- ❖ **Imtool()** is the inbuilt function in Matlab. It is used to display the image.
- ❖ It takes 2 parameters; the first is the image variable and the second is the range of intensity values.
- ❖ We provide an empty list as the second argument which means the complete range of intensity

**Morphological Operations**

Morphological operations are image-processing techniques used to analyze and process geometric structures in binary and grayscale images. These operations focus on the shape and structure of objects within an image.

Morphological operations aim to probe and transform an image based on its shape, enhancing features or removing imperfections.

Different Morphological Operations:

1. Erosion

Erosion is a fundamental morphological operation that reduces the size of objects in a binary image. It works by removing pixels from the boundaries of objects.

- ❖ **Purpose:** To remove small noise, detach connected objects, and erode boundaries.
- ❖ **How it Works:** The structuring element slides over the image, and for each position, if all the pixels under the structuring element match the foreground, the pixel in the output image is set to the foreground. Otherwise, it is set to the background.

2. Dilation

Dilation is the opposite of erosion and is used to increase the size of objects in an image.

- ❖ **Purpose:** To join adjacent objects, fill small holes, and enhance features.
- ❖ **How it Works:** The structuring element slides over the image, and for each position, if any pixel under the structuring element matches the foreground, the pixel in the output image is set to the foreground.

3. Opening

Opening is a compound operation that involves erosion followed by dilation.

- ❖ **Purpose:** To remove small objects or noise from the image while preserving the shape and size of larger objects.
- ❖ **How it Works:** First, the image undergoes erosion, which removes small objects and noise. Then, dilation is applied to restore the size of the remaining objects to their original dimensions.

4. Closing

Closing is another compound operation that consists of dilation followed by erosion.

- ❖ **Purpose:** To fill small holes and gaps in objects while preserving their overall shape.
- ❖ **How it Works:** First, dilation is applied to the image, filling small holes and gaps. Then, erosion is applied to restore the original size of the objects.

5. Hit-or-Miss Transform

The hit-or-miss transform is used to find specific patterns or shapes in a binary image.

- ❖ **Purpose:** To detect specific configurations or shapes in the image.
- ❖ **How it Works:** It uses a pair of structuring elements: one for the foreground and one for the background. The operation looks for places where the foreground structuring element matches the foreground in the image, and the background structuring element matches the background in the image.

6. Morphological Gradient

The morphological gradient is the difference between the dilation and erosion of an image.

- ❖ **Purpose:** To highlight the boundaries or edges of objects in the image.
- ❖ **How it Works:** By subtracting the eroded image from the dilated image, the boundaries of the objects are emphasized.

7. Top-Hat Transform

The top-hat transform is used to extract small elements and details from an image.

- ❖ **Purpose:** To enhance bright objects on a dark background or vice versa.
- ❖ **How it Works:** There are two types: white top-hat (original image minus the result of opening) and black top-hat (result of closing minus the original image).

8. Skeletonization

Skeletonization reduces objects in a binary image to their skeletal form.

- ❖ **Purpose:** To simplify objects to their essential structure while preserving their connectivity.
- ❖ **How it Works:** It iteratively erodes the image until the objects are reduced to a minimal, skeletal form.

9. Pruning

Pruning removes small spurs or extraneous branches from the skeleton of an image.

- ❖ **Purpose:** To clean up the skeleton by removing unwanted artifacts.
- ❖ **How it Works:** It identifies and removes small, irrelevant branches from the skeletonized image.

Implementing Morphological Operations in Python

```
import cv2
import numpy as np
from matplotlib import pyplot as plt

# Load an image
image = cv2.imread('/content/Screenshot-2024-06-11-150857.webp', 0) # Change
'input_image.jpg' to your image file path

# Define a kernel (structuring element)
kernel = np.ones((5,5), np.uint8)

# Erosion
erosion = cv2.erode(image, kernel, iterations=1)

# Dilation
dilation = cv2.dilate(image, kernel, iterations=1)

# Opening (erosion followed by dilation)
opening = cv2.morphologyEx(image, cv2.MORPH_OPEN, kernel)

# Closing (dilation followed by erosion)
closing = cv2.morphologyEx(image, cv2.MORPH_CLOSE, kernel)

# Opening (erosion followed by dilation)
opening = cv2.morphologyEx(image, cv2.MORPH_OPEN, kernel)

# Closing (dilation followed by erosion)
closing = cv2.morphologyEx(image, cv2.MORPH_CLOSE, kernel)

# Display the results
titles = ['Original Image', 'Erosion', 'Dilation', 'Opening', 'Closing']
images = [image, erosion, dilation, opening, closing]

for i in range(5):
    plt.subplot(2, 3, i+1)
    plt.imshow(images[i], cmap='gray')
    plt.title(titles[i])
    plt.xticks([], plt.yticks([]))

plt.show()
```

Output:

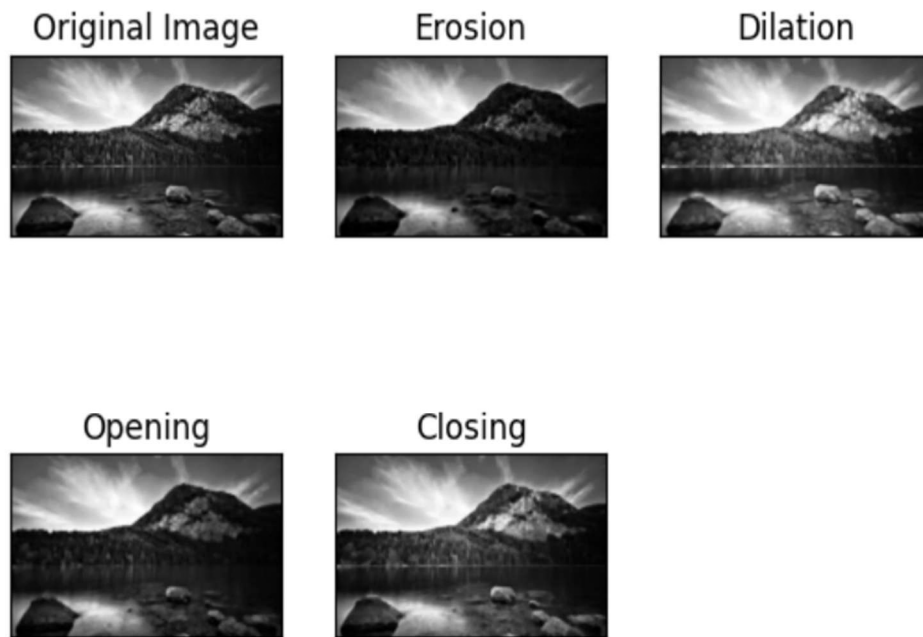


Image filtering

Filtering is the process of manipulating an image to alter its appearance. This can be done by applying special effects, such as blurring, sharpening, or color correction. Filters are used to modify the appearance of an image, making it look more interesting or realistic. They can also be used to create a certain effect, such as making an image look brighter, more vivid, or more detailed.

Types of Filters

There are several types of filters that can be used to modify an image. Some of the most common filters are a blur, sharpen, edge detection, color correction, and noise reduction.

Blur: Blur filters are used to soften the edges of an image, creating a more subtle look. This can be used to make an image look more natural or to reduce the visibility of distracting details.

Sharpen: Sharpen filters are used to make an image appear sharper and clearer. This can be used to make an image look more detailed or to make the colors more vibrant.

Edge Detection: Edge detection filters are used to accentuate the outlines of an image. This can be used to make objects stand out more or to create a more dramatic effect.

Color Correction: Color correction filters are used to adjust the hue, saturation, and brightness of an image. This can be used to make an image look more realistic, or to create a specific color palette.

Noise Reduction: Noise reduction filters are used to remove unwanted noise from an image. This can be used to make an image look more natural or to reduce the visibility of digital artifacts.

Applications of Filtering

Image processing: Filtering techniques can be used to enhance the clarity and detail of an image. For instance, a low-pass filter can be used to reduce noise, while a high-pass filter can be used to sharpen the image.

Video processing: Filtering techniques can be used to reduce noise and enhance the quality of a video. For instance, a low-pass filter can be used to reduce background noise, while a high-pass filter can be used to sharpen the video.

Medical imaging: Filtering techniques can be used to enhance the clarity and detail of medical images. For instance, a low-pass filter can be used to reduce noise and a high-pass filter can be used to sharpen the image.

3D rendering: Filtering techniques can be used to enhance the clarity and detail of 3D renderings. For instance, a low-pass filter can be used to reduce noise, while a high-pass filter can be used to sharpen the image.

Image Enhancement: Filtering is used to enhance the visual appearance of digital images. It can be used to improve contrast, sharpness, brightness, and color balance. It can also be used to reduce noise and artifacts.

Image Restoration: Filtering is also used for image restoration, which is the process of restoring a degraded image to its original state. This is often done by removing noise, sharpening the image, and correcting the color balance.

Image Compression: Filtering can be used to reduce the size of an image without compromising on its quality. This is achieved by eliminating redundant information in the image.

Image Segmentation: Filtering can also be used for image segmentation, which is the process of separating an image into regions of interest. This is often done by applying a filter to the image to identify regions with specific characteristics.

Thresholding

Thresholding is one of the segmentation techniques that generates a binary image (a binary image is one whose pixels have only two values - 0 and 1 and thus requires only one bit to store pixel intensity) from a given grayscale image by separating it into two regions based on a threshold value. Hence pixels having intensity values greater than the said threshold will be treated as white or 1 in the output image and the others will be black or 0.

When the intensity distribution of objects and background are sufficiently distinct, it is possible to use a single or global threshold applicable over the entire image. The basic global thresholding algorithm iteratively finds the best threshold value so segmenting.

Step 1: Import libraries and Image Preparation

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

image = cv2.imread('input1.jpg')
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

Step 2: Helper Function, Define the helper function which helps in displaying the images

```
def show_image(img, title):
    plt.imshow(img, cmap='gray')
    plt.title(title)
    plt.axis('off')
    plt.show()
```

Step 3: Display Original Image

```
show_image(gray_image, "Original Grayscale Image")
```

Output:

Original Grayscale Image



Step 4: Otsu's Thresholding: The threshold value is not provided by us, instead, Otsu's method determines it automatically based on the image's histogram. This makes separation of foreground and background particularly strong on bimodal images.

```
ret, otsu_thresh = cv2.threshold(  
    gray_image, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)  
print("Calculated Otsu threshold value:", ret)  
show_image(otsu_thresh, "Otsu's Thresholding")
```

Otsu's Thresholding



Overview of OpenCV and Vision tools

OpenCV is a cross-platform library using which we can develop real-time computer vision applications. It mainly focuses on image processing, video capture and analysis including features like face detection and object detection.

Brief History of OpenCV

OpenCV was initially an Intel research initiative to advise CPU-intensive applications. It was officially launched in 1999.

- ❖ In the year 2006, its first major version, OpenCV 1.0 was released.
- ❖ In October 2009, the second major version, OpenCV 2 was released.
- ❖ In August 2012, OpenCV was taken by a nonprofit organization OpenCV.org.

How Computers See Images

- ❖ **Pixel Values:** Each image is made up of pixels and each pixel holds values for color and brightness. In color images, each pixel generally has three values (Red, Green, Blue).
- ❖ **Matrix Representation:** The image is stored as a NumPy array, where each entry shows a pixel's intensity. Grayscale images are 2D arrays and color images are 3D arrays.
- ❖ **Processing:** Algorithms manipulate these arrays to apply transformations like edge detection, blurring or object detection.

Setting Up OpenCV in Python

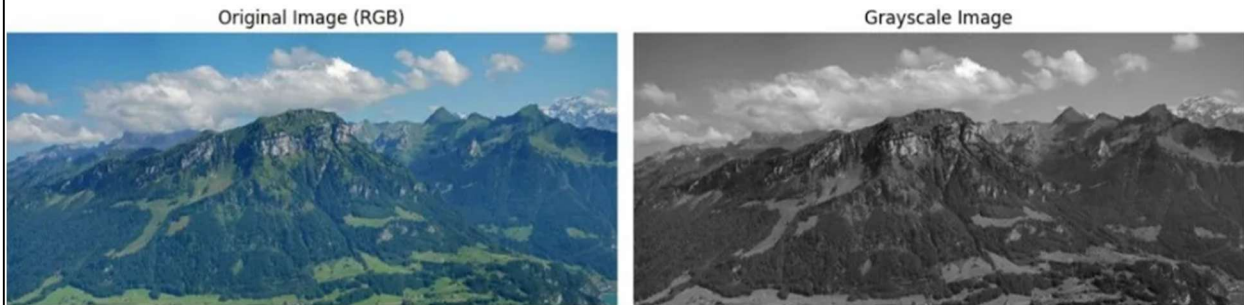
Importing Libraries

- ❖ **cv2:** For image reading and processing.
- ❖ **numpy:** For image array operations (used internally).
- ❖ **matplotlib.pyplot:** For displaying the image visually.
- ❖ **os:** For potential path operations (though not directly used here).

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os
image_path = '/content/Sample_CV.jpg'
img = cv2.imread(image_path)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
img_gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.imshow(img_rgb)
plt.title("Original Image (RGB)")
plt.axis("off")
plt.subplot(1, 2, 2)
plt.imshow(img_gray, cmap='gray')
```



```
plt.title("Grayscale Image")  
plt.axis("off")  
plt.tight_layout()  
plt.show()
```



Functionalities of OpenCV

- ❖ **Image and Video I/O:** Read, display and save images or video files.
- ❖ **Image Processing:** Perform operations like resizing, filtering, thresholding and color transformations.
- ❖ **Object Detection:** Detect faces, shapes and features using built-in classifiers and detectors.
- ❖ **Machine Learning:** Train and apply models for classification and clustering.
- ❖ **Computational Photography:** Apply advanced effects like denoising and panorama stitching.

Computer Vision Tools

- ❖ **BoofCV** covering all the basic image processing capabilities like low-level image processing, structure from motion, recognition, fiducial detection, feature detection and tracking.
- ❖ **OpenCV** used open-source libraries when it comes to computer vision and image processing.
- ❖ **TensorFlow** can be used as a free resource, which is mainly involved in deep learning applications while traditional machine learning methods.
- ❖ **Matlab** is used to create algorithms and further the exploration of data with efficient vision solutions.
- ❖ **NVIDIA** can contribute to the increased speed of development in computer vision applications.
- ❖ **SimpleCV** is developers to build computer vision apps and with the support of high-powered computer vision libraries such as OpenCV
- ❖ **Keras** is developers to build computer vision apps and with the support of high-powered computer vision libraries such as OpenCV

- ❖ **Theano** is multifunctional tool that is capable of running on both CPU and GPU architectures.
- ❖ **YOLO**, which is “You Only Look Once,” The algorithm performs object detection faster than any other algorithm.
- ❖ **CAFFE** is a deep-learning library that is very popular for its fast processing and modulatory capability.

Ethical Concerns and Challenges in Computer Vision

- ❖ **Data Privacy:** Images and videos captured by systems are collected, stored, processed, and shared securely and ethically.
- ❖ **Bias:** AI model's predictions are influenced by imbalanced or unrepresentative image data.
- ❖ **Surveillance:** Surveillance in computer vision involves automated monitoring of environments using image and video analysis techniques.
- ❖ **Explainability:** It make image-based AI decisions understandable to humans.
- ❖ **Safety:** Computer vision-based AI systems operate reliably without causing harm.
- ❖ **Accountability:** Assigning responsibility and ensuring transparency for AI-driven visual decisions.
- ❖ **Security:** Assigning responsibility and ensuring transparency for AI-driven visual decisions.
- ❖ **Misuse:** It is harmful or unethical application of vision-based AI systems.

Difference between human vision & Computer (Machine) Vision

Task	Human vision	Machine vision
Visualization	Passive, mainly by reflection of light from opaque surfaces	Passive and active (controlled illumination) using electromagnetic, particulate, and acoustic radiation
Image formation	Refractive optical system	Various systems
Control of irradiance	Muscle-controlled pupil	Motorized apertures, filter wheels, tunable filters
Focusing	Muscle-controlled change of focal length	Autofocus systems based on various principles of distance measurements
Irradiance resolution	Logarithmic sensitivity	Linear sensitivity, quantization between 8- and 16-bits; logarithmic sensitivity, for example, HDRC-sensors
Tracking	Highly mobile eyeball	Scanner and robot-mounted cameras
Processing and analysis	Hierarchically organized massively parallel processing	Serial processing still dominant; parallel processing not in general use

S.No	Questions
	Define image processing? Explain different basic steps involved in digital image processing.
01	Define image? Explain different types of images.
02	Explain Phases of Image Processing.
03	List and explain Advantages & Disadvantages of Digital Image Processing.
04	Write short note on 1) Histogram 2) Probability Mass Function (PMF) 3) Cumulative Distributive Function(CDF) 4) Convolution
05	Explain Edge detection with suitable code.
06	Define Morphological Operations? and explain different types of Morphological Operations with suitable example.
07	Write code in Python Morphological Operations
08	Define Image filtering? Mention Advantages & Disadvantages of Image filtering.
09	Explain types of Filters used in Image filtering.
10	Define Thresholding, Write code for Otsu's Thresholding.
11	Define openCV? Explain how Computers See Images
12	Explain functionalities of OpenCV
13	List and explain Computer Vision Tools
14	Explain ethical Concerns and Challenges in Computer Vision
15	List difference between human vision & Computer (Machine) Vision